STA 2002, Summer 2024
Probability and Statistics II

Distribution-free Confidence Intervals
Sign tests and Wilcoxon tests

Recap: CI for normal mean

- For mean(median) μ from a normal population with known σ,

$$X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2) \rightarrow \bar{X} \sim N(\mu, \frac{\sigma^2}{n}) \quad \Rightarrow \quad \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

$$P\left\{ -z_{\alpha/2} \leq \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2} \right\} = 1 - \alpha$$

Confidence level

$$\bar{x} - Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

This CI can be used for mean of non-normal distributions when *n*>30 (CLT)

Q: What if the distribution is not normal or unknown?
How can we construct CI for the **median**?

Distribution-free CI for Percentiles

- Not assuming normal distribution
- No requirement on the sample size

Estimate the population(theoretical) percentiles/quantiles:

- The p-th **theoretical percentile** of the distribution F is $\boldsymbol{\pi_p}$:

$$F(\pi_p) = p \qquad \pi_p = F^{-1}(p)$$

- Example: $\pi_{0.5}$ is called the **median** of the distribution F.

- $X_{(k)}, k = (n+1)p$ is the p-th **sample quantile** $\hat{\boldsymbol{\pi}}_{\boldsymbol{p}}$.

How to construct a confidence interval for π_p?

Reading: Chapter 7.5

CI for Median

Let $X_1, X_2, \dots, X_n$ be i.i.d. random samples from a common distribution F.

$$X_1, X_2, \dots, X_n \sim F$$

The **median $m = \pi_{1/2}$**: $\boxed{F(m) = 1/2}$

$$P(X_i \le m) = 1/2$$

Ordered statistics $X_{(1)} < X_{(2)} < \cdots < X_{(n)}$ ⟹ ***CI***: $(X_{(1)}, X_{(n)})$?

Confidence level $P(X_{(1)} < m < X_{(n)}) = 1 - P(X_{(1)} \ge m) - P(X_{(n)} \le m)$

$$= 1 - (1 - F(m)^n - F(m)^n$$

$$= 1 - \frac{1}{2^n} - \frac{1}{2^n} = 1 - \frac{1}{2^{n-1}}$$

CI for Median

Order statistics $X_{(1)} < X_{(2)} < \cdots < X_{(n)}$ ⟹ ***CI***: $(X_{(i)}, X_{(j)})$?

$$W = \sum 1\{X_i \le m\}$$
$$W \sim Bin\left(n, \frac{1}{2}\right)$$

Confidence level of $(X_{(i)}, X_{(j)})$

$$P(X_{(i)} < m < X_{(j)}) = P(i \le W \le j-1)$$

$$= \sum_{l=i}^{j-1} \binom{n}{l} \left(\frac{1}{2}\right)^{l} \left(1-\frac{1}{2}\right)^{n-l}$$

$$= \sum_{l=i}^{j-1} \binom{n}{l} \left(\frac{1}{2}\right)^{n}$$

Example

Suppose we have samples

$$x_1 = 8.7, x_2 = 0.1, x_3 = 4.4, x_4 = 3.1, x_5 = 6.0$$

$$0.1 < 3.1 < 4.4 < 6.0 < 8.7$$

CI: (0.1,8.7) ***Confidence level*** $= 1 - \frac{1}{2^{n-1}} = \frac{15}{16} = 94\%$

CI: (3.1,6.0) ***Confidence level*** $= \left(\binom{5}{2} + \binom{5}{3}\right) \times \left(\frac{1}{2}\right)^5 = 62.5\%$

CI: (3.1,8.7) ***Confidence level*** $= \left(\binom{5}{2} + \binom{5}{3} + \binom{5}{4}\right) \times \left(\frac{1}{2}\right)^5 = 78\%$

……

A Useful Table

Table 7.5-1 Information for confidence intervals for m

n	$(i, n+1-i)$	$P(Y_i < m < Y_{n+1-i})$	n	$(i, n+1-i)$	$P(Y_i < m < Y_{n+1-i})$
5	(1, 5)	0.9376	13	(3, 11)	0.9776
6	(1, 6)	0.9688	14	(4, 11)	0.9426
7	(1, 7)	0.9844	15	(4, 12)	0.9648
8	(2, 7)	0.9296	16	(5, 12)	0.9232
9	(2, 8)	0.9610	17	(5, 13)	0.9510
10	(2, 9)	0.9786	18	(5, 14)	0.9692
11	(3, 9)	0.9346	19	(6, 14)	0.9364
12	(3, 10)	0.9614	20	(6, 15)	0.9586

close to 95%

Example

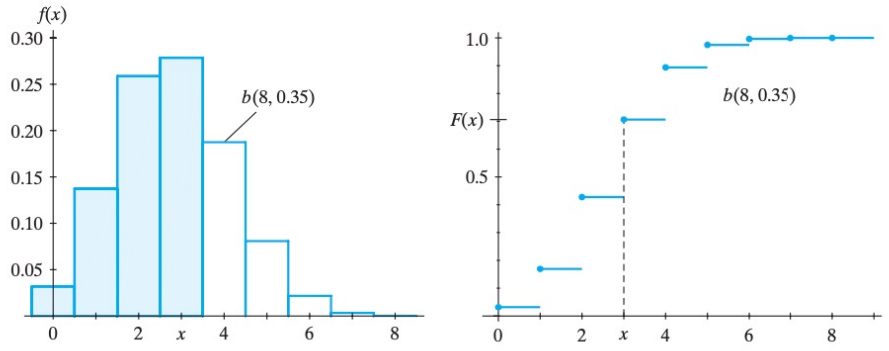
Example 7.5-1 The lengths in centimeters of $n = 9$ fish of a particular species captured off the New England coast were 32.5, 27.6, 29.3, 30.1, 15.5, 21.7, 22.8, 21.2, and 19.0. Thus, the observed order statistics are

$$15.5 < 19.0 < 21.2 < 21.7 < 22.8 < 27.6 < 29.3 < 30.1 < 32.5.$$

CI: (19,30.1) ***Confidence level*** $= \sum_{k=2}^{7} \binom{9}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{9-k} = 0.9610$

n	$(i, n+1-i)$	$P(Y_i < m < Y_{n+1-i})$
9	(2, 8)	0.9610

Table II The Binomial Distribution



$$F(x) = P(X \le x) = \sum_{k=0}^{x} \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

		p									
n	x	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
2	0	0.9025	0.8100	0.7225	0.6400	0.5625	0.4900	0.4225	0.3600	0.3025	0.2500
	1	0.9975	0.9900	0.9775	0.9600	0.9375	0.9100	0.8775	0.8400	0.7975	0.7500
	2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
3	0	0.8574	0.7290	0.6141	0.5120	0.4219	0.3430	0.2746	0.2160	0.1664	0.1250
	1	0.9928	0.9720	0.9392	0.8960	0.8438	0.7840	0.7182	0.6480	0.5748	0.5000
	2	0.9999	0.9990	0.9966	0.9920	0.9844	0.9730	0.9571	0.9360	0.9089	0.8750
	3	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
4	0	0.8145	0.6561	0.5220	0.4096	0.3164	0.2401	0.1785	0.1296	0.0915	0.0625
	1	0.9860	0.9477	0.8905	0.8192	0.7383	0.6517	0.5630	0.4752	0.3910	0.3125
	2	0.9995	0.9963	0.9880	0.9728	0.9492	0.9163	0.8735	0.8208	0.7585	0.6875
	3	1.0000	0.9999	0.9995	0.9984	0.9961	0.9919	0.9850	0.9744	0.9590	0.9375
	4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table II *continued*

		p									
n	x	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
	2	0.9962	0.9743	0.9262	0.8520	0.7564	0.6471	0.5323	0.4199	0.3164	0.2266
	3	0.9998	0.9973	0.9879	0.9667	0.9294	0.8740	0.8002	0.7102	0.6083	0.5000
	4	1.0000	0.9998	0.9988	0.9953	0.9871	0.9712	0.9444	0.9037	0.8471	0.7734
	5	1.0000	1.0000	0.9999	0.9996	0.9987	0.9962	0.9910	0.9812	0.9643	0.9375
	6	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9994	0.9984	0.9963	0.9922
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
8	0	0.6634	0.4305	0.2725	0.1678	0.1001	0.0576	0.0319	0.0168	0.0084	0.0039
	1	0.9428	0.8131	0.6572	0.5033	0.3671	0.2553	0.1691	0.1064	0.0632	0.0352
	2	0.9942	0.9619	0.8948	0.7969	0.6785	0.5518	0.4278	0.3154	0.2201	0.1445
	3	0.9996	0.9950	0.9786	0.9437	0.8862	0.8059	0.7064	0.5941	0.4770	0.3633
	4	1.0000	0.9996	0.9971	0.9896	0.9727	0.9420	0.8939	0.8263	0.7396	0.6367
	5	1.0000	1.0000	0.9998	0.9988	0.9958	0.9887	0.9747	0.9502	0.9115	0.8555
	6	1.0000	1.0000	1.0000	0.9999	0.9996	0.9987	0.9964	0.9915	0.9819	0.9648
	7	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9993	0.9983	0.9961
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
9	0	0.6302	0.3874	0.2316	0.1342	0.0751	0.0404	0.0207	0.0101	0.0046	0.0020
	1	0.9288	0.7748	0.5995	0.4362	0.3003	0.1960	0.1211	0.0705	0.0385	0.0195
	2	0.9916	0.9470	0.8591	0.7382	0.6007	0.4628	0.3373	0.2318	0.1495	0.0898
	3	0.9994	0.9917	0.9661	0.9144	0.8343	0.7297	0.6089	0.4826	0.3614	0.2539
	4	1.0000	0.9991	0.9944	0.9804	0.9511	0.9012	0.8283	0.7334	0.6214	0.5000
	5	1.0000	0.9999	0.9994	0.9969	0.9900	0.9747	0.9464	0.9006	0.8342	0.7461
	6	1.0000	1.0000	1.0000	0.9997	0.9987	0.9957	0.9888	0.9750	0.9502	0.9102
	7	1.0000	1.0000	1.0000	1.0000	0.9999	0.9996	0.9986	0.9962	0.9909	0.9805
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9992	0.9980
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

More in Table II, Appendix B

CLT Approximation When *n* is Large

CLT Approximation for Large n

$$W = \sum 1\{X_i \le m\}$$

$$W \sim Bin\left(n, \frac{1}{2}\right)$$

CLT ⇒

$$W \sim N\left(\frac{n}{2}, \frac{n}{4}\right)$$

Required

Confidence level of $(X_{(i)}, X_{(j)})$

$$P(X_{(i)} < m < X_{(j)}) = P(i \le W \le j-1)$$

$$= P\left(\frac{i - n/2}{\sqrt{n/4}} \le \underbrace{\frac{W - n/2}{\sqrt{n/4}}}_{N(0,1)} \le \frac{j-1-n/2}{\sqrt{n/4}}\right)$$

$$\approx \Phi\left(\frac{j-1-n/2}{\sqrt{n/4}}\right) - \Phi\left(\frac{i-n/2}{\sqrt{n/4}}\right)$$

With Half-unit continuity correction

$$P(X_{(i)} < m < X_{(j)}) = P(i \le W \le j-1)$$

$$= P(i - 1/2 \le W \le j - 1/2)$$

$$= P\left(\frac{i-1/2-n/2}{\sqrt{n/4}} \le \frac{\overset{\vee}{W} - n/2}{\sqrt{n/4}} \le \frac{j-1/2-n/2}{\sqrt{n/4}}\right)$$

$$\approx \Phi\left(\frac{j-1/2-n/2}{\sqrt{n/4}}\right) - \Phi\left(\frac{i-1/2-n/2}{\sqrt{n/4}}\right)$$

half-unit correction for continuity

$$W \sim Bin(n,p) \xrightarrow{\text{CLT}} W \sim N\big(np, np(1-p)\big) \Rightarrow \frac{W-np}{\sqrt{np(1-p)}} \sim N(0,1)$$

The "pdf" for W

$$f(y) = \frac{n!}{k!(n-k)!}p^k(1-p)^{n-k} \qquad k-1/2 < y = k < k+1/2$$
$$k = 0,1,2,\ldots,n.$$

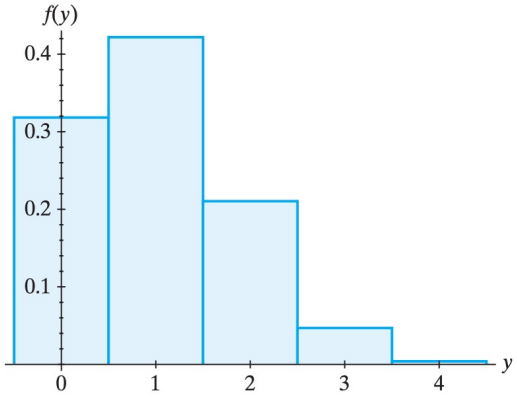


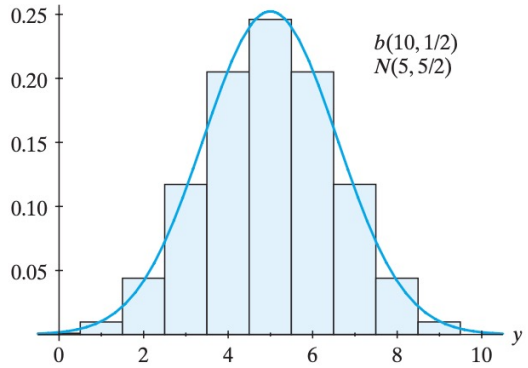
Figure 5.7-1 Probability histogram for $b(4,1/4)$

$$\Rightarrow P(k-1/2 \le W \le k+1/2) = P(W=k) = \frac{n!}{k!(n-k)!}p^k(1-p)^{n-k}$$

Half-unit continuity correction

$$P(i \le W \le j) = P(i-1/2 \le W \le j+1/2)$$

$$= P\left(\frac{i-1/2-np}{\sqrt{np(1-p)}} \le \frac{W-np}{\sqrt{np(1-p)}} \le \frac{j+1/2-np}{\sqrt{np(1-p)}}\right)$$
$$= \Phi\left(\frac{j+1/2-np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{i-1/2-np}{\sqrt{np(1-p)}}\right)$$



Example

Suppose now the sample size is n = 27. What is the confidence level of the confidence interval $(X_{(10)}, X_{(18)})$ for the median m?

$$W \sim Bin\left(n, \frac{1}{2}\right) \xrightarrow{\text{CLT}} W \sim N\left(\frac{27}{2}, \frac{27}{4}\right)$$

Confidence level of $(X_{(10)}, X_{(18)})$

$$P(X_{(10)} < m < X_{(18)}) = P(10 \le W \le 17)$$

$$= P(9.5 \le W \le 17.5)$$ **With half-unit continuity correction**

$$= P\left(\frac{9.5 - 13.5}{\sqrt{27/4}} \le \overbrace{\frac{W - 13.5}{\sqrt{27/4}}}^{N(0,1)} \le \frac{17.5 - 13.5}{\sqrt{27/4}}\right)$$

$$\approx \Phi(1.54) - \Phi(-1.54) = 0.8764$$

CI for any Percentiles π_p

- The p-th **theoretical percentile** of the distribution F is $\boldsymbol{\pi_p}$:

$$F(\pi_p) = p$$

$$W = \sum 1\{X_i \leq \pi_p\}$$
$$W \sim Bin(n, p)$$

Confidence level of $(X_{(i)}, X_{(j)})$

$$P(X_{(i)} < \pi_p < X_{(j)}) = P(i \leq W \leq j-1)$$
$$= \sum_{l=i}^{j-1} \binom{n}{l} p^l (1-p)^{n-l}$$

CLT approximation

- The p-th **theoretical percentile** of the distribution F is $\boldsymbol{\pi_p}$:

$$F(\pi_p) = p$$

$$W = \sum 1\{X_i \le \pi_p\}$$
$$W \sim Bin(n, p)$$

CLT $\Rightarrow$ $W \sim N\big(np, np(1-p)\big)$

Confidence level of $(X_{(i)}, X_{(j)})$

With Half-unit continuity correction

$$P(X_{(i)} < \pi_p < X_{(j)}) = P(i \le W \le j-1) = P(i - 1/2 \le W \le j - 1/2)$$

$$= P\left(\frac{i - 1/2 - np}{\sqrt{np(1-p)}} \le \frac{W - np}{\sqrt{np(1-p)}} \le \frac{j - 1/2 - np}{\sqrt{np(1-p)}}\right)$$

($\frac{W - np}{\sqrt{np(1-p)}} \sim N(0,1)$)

$$\approx \Phi\left(\frac{j - 1/2 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{i - 1/2 - np}{\sqrt{np(1-p)}}\right)$$

Example

Suppose now n = 25, and we would like to estimate $\pi_{0.25}$.

A point estimate: the p-th (p=0.25) sample quantile

- Order the samples $X_{(1)} < X_{(2)} < \cdots < X_{(n)}$
- $r = (n+1)p = \frac{n+1}{4} = 6.5$
- r is not an integer, $r = 6 + 0.5$, the sample quantile is

$$\hat{\pi}_{0.25} = X_{(6)} + 0.5(X_{(7)} - X_{(6)}) = (X_{(6)} + X_{(7)})/2$$

$$W = \sum 1\{X_i \le \pi_{0.25}\}$$

$$W \sim Bin(25, 0.25)$$

CI: $((X_{(3)}, X_{(10)})$

Confidence level

$$= \sum_{k=3}^{9} \binom{25}{k} (0.25)^k (0.75)^{25-k} = 0.8966$$

(Exact value, from Table II, Appendix B)

$$\approx \Phi\left(\frac{9.5 - 25/4}{\sqrt{75/16}}\right) - \Phi\left(\frac{2.5 - 25/4}{\sqrt{75/16}}\right) = 0.8917$$

(Approximated value, from CLT with continuity correction) **More accurate**

As a comparison: $\Phi\left(\frac{9-25/4}{\sqrt{75/16}}\right) - \Phi\left(\frac{3-25/4}{\sqrt{75/16}}\right) = 0.8313$ (Approximated value, from CLT, without half-unit continuity correction)

Example

Let the following numbers represent the order statistics of the $n = 27$ observations obtained in a random sample from a certain population of incomes (measured in hundreds of dollars):

261	269	271	274	279	280	283	284	286
287	292	293	296	300	304	305	313	321
322	329	341	343	356	364	391	417	476

Say we are interested in estimating the 25th percentile, $\pi_{0.25}$, of the population.

$$W = \sum 1\{X_i \le \pi_{0.25}\}$$

$$W \sim Bin(27, 0.25)$$

A point estimate: the p-th (p=0.25) sample quantile

- $r = (n+1)p = 7$
- r is an integer, the sample quantile is $\hat{\pi}_{0.25} = X_{(7)} = 283$

CI: $((X_{(4)}, X_{(10)})$ ***Confidence level*** $= \sum_{k=4}^{9} \binom{27}{k} (0.25)^k (0.75)^{27-k} = 0.8201$

Normal approximation with correction $\approx \Phi\left(\frac{9.5 - 27/4}{\sqrt{81/16}}\right) - \Phi\left(\frac{3.5 - 27/4}{\sqrt{81/16}}\right) = 0.8149$ **More accurate**

Normal approximation without correction $\approx \Phi\left(\frac{9 - 27/4}{\sqrt{81/16}}\right) - \Phi\left(\frac{4 - 27/4}{\sqrt{81/16}}\right) = 0.7305$

poor

Advantages of Distribution-free CI

- Little assumptions: all we assume is that the distribution is of the continuous type
- Works better and more robust for distributions deviate a lot from normal (highly skewed or heavy−tailed)
- Can be used to get confidence intervals for various percentiles, not just the median

Distribution-free Hypothesis Test

- Sign test
- Wilcoxon test

Reading: Chapter 8.4

Test Median

- Interested in testing the median m of a distribution
- Observe samples $X_1, X_2, \ldots, X_n$ from this distribution
- Deciding whether or not the median is m_0

Null hypothesis $H_0: m = m_0$

Alternative hypothesis $H_1: m \neq m_0$

Two-sided Hypothesis Test

$H_1: m > m_0$

$H_1: m < m_0$

One-sided Hypothesis Test

Sign test for median

$X_1, \ldots, X_n \sim F$ $\quad$ $\boxed{F(m) = 1/2}$ $\quad\Rightarrow\quad$ $P(X_i \leq m) = 1/2$

Null Hypothesis

$H_0: m = m_0$

Test Statistic

$W = \sum 1\{X_i \leq m_0\}$

Distribution under H_0

$W \sim Bin\left(n, \frac{1}{2}\right)$

Idea: Reject null hypothesis H_0 when *W* significantly differs from the Binomial distribution $Bin\left(n, \frac{1}{2}\right)$

Example

Example 8.4-1

Let X denote the length of time in seconds between two calls entering a call center. Let m be the unique median of this continuous-type distribution. We test the null hypothesis H_0: $m = 6.2$ against the alternative hypothesis H_1: $m < 6.2$. Table II in Appendix B tells us that if Y is the number of lengths of time between calls in a random sample of size 20 that are less than 6.2, then the critical region $C = \{y : y \geq 14\}$ has a significance level of $\alpha = 0.0577$. A random sample of size 20 yielded the following data:

6.8	5.7	6.9	5.3	4.1	9.8	1.7	7.0
2.1	19.0	18.9	16.9	10.4	44.1	2.9	2.4
4.8	18.9	4.8	7.9				

Since $y = 9$, the null hypothesis is not rejected. ■

$$\alpha = P(y \geq 14 | y \sim Bin(20,0.5)) = \sum_{l=14}^{20} \binom{n}{l} (0.5)^l (0.5)^{n-l} = 0.0577$$

Example

20	0	0.3585	0.1216	0.0388	0.0115	0.0032	0.0008	0.0002	0.0000	0.0000	0.0000
	1	0.7358	0.3917	0.1756	0.0692	0.0243	0.0076	0.0021	0.0005	0.0001	0.0000
	2	0.9245	0.6769	0.4049	0.2061	0.0913	0.0355	0.0121	0.0036	0.0009	0.0002
	3	0.9841	0.8670	0.6477	0.4114	0.2252	0.1071	0.0444	0.0160	0.0049	0.0013
	4	0.9974	0.9568	0.8298	0.6296	0.4148	0.2375	0.1182	0.0510	0.0189	0.0059

Table II *continued*

		p									
n	*x*	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
	5	0.9997	0.9887	0.9327	0.8042	0.6172	0.4164	0.2454	0.1256	0.0553	0.0207
	6	1.0000	0.9976	0.9781	0.9133	0.7858	0.6080	0.4166	0.2500	0.1299	0.0577
	7	1.0000	0.9996	0.9941	0.9679	0.8982	0.7723	0.6010	0.4159	0.2520	0.1316
	8	1.0000	0.9999	0.9987	0.9900	0.9591	0.8867	0.7624	0.5956	0.4143	0.2517
	9	1.0000	1.0000	0.9998	0.9974	0.9861	0.9520	0.8782	0.7553	0.5914	0.4119
	10	1.0000	1.0000	1.0000	0.9994	0.9961	0.9829	0.9468	0.8725	0.7507	0.5881
	11	1.0000	1.0000	1.0000	0.9999	0.9991	0.9949	0.9804	0.9435	0.8692	0.7483
	12	1.0000	1.0000	1.0000	1.0000	0.9998	0.9987	0.9940	0.9790	0.9420	0.8684
	13	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9985	0.9935	0.9786	0.9423
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9984	0.9936	0.9793
	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9985	0.9941
	16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9997	0.9987
	17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998
	18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	19	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Example

For the same dataset, say we still want to test

$$H_0: m = 6.2 \text{ against the alternative hypothesis } H_1: m < 6.2.$$

What is the p-value of the test?

6.8	5.7	6.9	5.3	4.1	9.8	1.7	7.0
2.1	19.0	18.9	16.9	10.4	44.1	2.9	2.4
4.8	18.9	4.8	7.9				

$y = 9$

Large value of *y* favors H_1

p-value $= P(y \geq 9 | y \sim Bin(20,0.5)) = \sum_{l=9}^{20} \binom{n}{l} (0.5)^l (0.5)^{n-l} = 0.7483 \gg 0.05$

p-value > significance level, we fail to reject H_0.

Example

Again for the same dataset, say we want to test

$$H_0 : m = 7.2, \quad H_1 : m \neq 7.2.$$

For such two-sided test, we construct a two-sided rejection region C:

Reject H_0 when $y \geq 15$ or $y \leq 5$

significance level

$$\alpha = P(y \geq 15 \text{ or } y \leq 5 | y \sim Bin(20,0.5)) = 1 - \sum_{l=6}^{14} \binom{n}{l} (0.5)^l (0.5)^{n-l} = 0.0414$$

6.8	5.7	6.9	5.3	4.1	9.8	1.7	7.0
2.1	19.0	18.9	16.9	10.4	44.1	2.9	2.4
4.8	18.9	4.8	7.9				

$y = 12$

fail to reject H_0.

p-value $= 2P(y \geq 12 | y \sim Bin(20,0.5)) = 0.5034.$ ⇨ fail to reject H_0.

Wilcoxon Test

$X_1, \ldots, X_n \sim F$ $\quad$ $F(m) = 1/2$

Null Hypothesis

$H_0: m = m_0$

Wilcoxon test procedure

Magnitudes (not used in sign test)

1) Rank the absolute values $|X_1 - m_0|, |X_2 - m_0|, \ldots, |X_n - m_0|$, and denote the rank of $|X_k - m_0|$ as R_k.
2) With each R_k, we calculate the **signed ranks**
   - If $X_k - m_0 > 0$, we use R_k
   - If $X_k - m_0 < 0$, we use $-R_k$
3) The Wilcoxon signed rank statistic

$$W = \sum_{k=1}^{n} \text{signed ranks}$$

Example

Example 8.4-2

Suppose the lengths of $n = 10$ sunfish are

$$x_i: \ 5.0 \ 3.9 \ 5.2 \ 5.5 \ 2.8 \ 6.1 \ 6.4 \ 2.6 \ 1.7 \ 4.3$$

We shall test $H_0: m = 3.7$ against the alternative hypothesis $H_1: m > 3.7$. Thus, we have

$x_k - m_0$:	1.3,	0.2,	1.5,	1.8,	−0.9,	2.4,	2.7,	−1.1,	−2.0,	0.6
$\lvert x_k - m_0 \rvert$:	1.3,	0.2,	1.5,	1.8,	0.9,	2.4,	2.7,	1.1,	2.0,	0.6
Ranks:	5,	1,	6,	7,	3,	9,	10,	4,	8,	2
Signed Ranks:	5,	1,	6,	7,	−3,	9,	10,	−4,	−8,	2

Therefore, the Wilcoxon statistic is equal to

$$W = 5 + 1 + 6 + 7 - 3 + 9 + 10 - 4 - 8 + 2 = 25.$$

Distribution of W Under H_0

- Assume the random variable X is of the continuous type
- Assume that the pdf of X is **symmetric** about the median

Null Hypothesis

$$X_1, \ldots, X_n \sim F \qquad H_0: m = m_0 \qquad W = \sum_{k=1}^{n} \text{signed ranks}$$

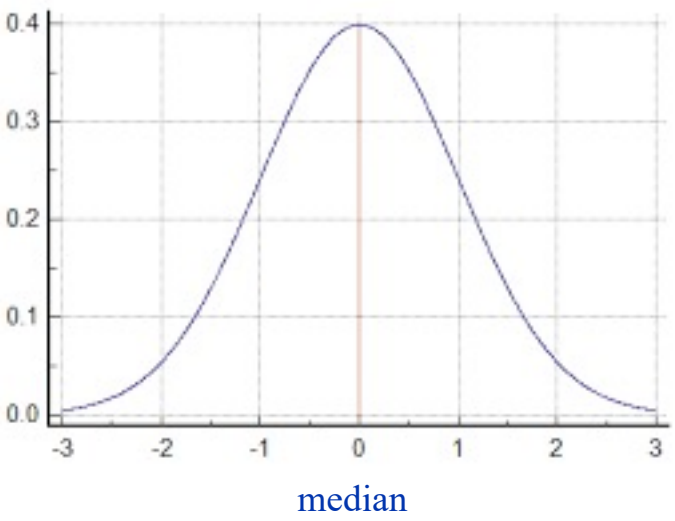
Signed ranks: Under Null Hypothesis $F(m_0) = P(X \le m_0) = 1/2$

- If $X_k - m_0 > 0$, we use R_k (**happens with probability 0.5**)
- If $X_k - m_0 < 0$, we use $-R_k$ (**happens with probability 0.5**)

$R_1, R_2, \ldots, R_n$ is just a permutation of $1,2, \ldots, n$.

Under H_0

$$P(X_k < m_0) = P(X_k > m_0) = \frac{1}{2}, \qquad k = 1, 2, \ldots, n.$$



Under H_0, W is a sum that contains $1,2, \ldots, n$, each with a positive (with probability 0.5) or a negative sign (with probability 0.5).

Distribution of W Under H_0

- Assume the random variable X is of the continuous type
- Assume that the pdf of X is **symmetric** about the median

Null Hypothesis

$$X_1, \ldots, X_n \sim F \qquad H_0: m = m_0 \qquad W = \sum_{k=1}^{n} \text{signed ranks}$$

Under H_0, W is a sum that contains $1,2,\ldots,n$, each with a positive (with probability 0.5) or a negative sign (with probability 0.5).

➡ ***W* has the same distribution as *V*.**

$$V = \sum_{k=1}^{n} V_k \qquad P(V_k = k) = P(V_k = -k) = \frac{1}{2}, \qquad k = 1, 2, \ldots, n.$$

$$E(V_k) = -k\left(\frac{1}{2}\right) + k\left(\frac{1}{2}\right) = 0$$

$$\text{Var}(V_k) = E(V_k^2) = (-k)^2\left(\frac{1}{2}\right) + (k)^2\left(\frac{1}{2}\right) = k^2$$

Distribution of W Under H_0

- Assume the random variable X is of the continuous type
- Assume that the pdf of X is **symmetric** about the median

Null Hypothesis

$$X_1, \ldots, X_n \sim F \qquad H_0: m = m_0 \qquad W = \sum_{k=1}^{n} \text{signed ranks}$$

Under H_0

$$E(W) = E(V) = \sum_{k=1}^{n} E(V_k) = 0.$$

$$\text{Var}(W) = \text{Var}(V) = \sum_{k=1}^{n} \text{Var}(V_k) = \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

The exact distribution of *W*

MGF of *W (V)*

$$M_{V_k}(t) = e^{-kt} \times \frac{1}{2} + e^{kt} \times \frac{1}{2} = \frac{e^{-kt} + e^{kt}}{2}$$

Examples

n=2:

$$M_V(t) = M_{V_1+V_2}(t) = \left(\frac{e^{-t} + e^{t}}{2}\right)\left(\frac{e^{-2t} + e^{2t}}{2}\right) = \frac{e^{-3t} + e^{-t} + e^{t} + e^{3t}}{4}$$

W	-3	-1	1	3
probability	1/4	1/4	1/4	1/4

n=3:

$$M_V(t) = M_{V_1+V_2+V_3}(t) = \left(\frac{e^{-t} + e^{t}}{2}\right)\left(\frac{e^{-2t} + e^{2t}}{2}\right)\left(\frac{e^{-3t} + e^{3t}}{2}\right) = \frac{e^{-6t} + e^{-4t} + e^{-2t} + 2e^{0} + e^{2t} + e^{4t} + e^{6t}}{8}$$

W	-6	-4	-2	0	2	4	6
probability	1/8	1/8	1/8	2/8	1/8	1/8	1/8

……

CLT Approximation of W Under H_0

- Assume the random variable X is of the continuous type
- Assume that the pdf of X is **symmetric** about the median

$$X_1, \ldots, X_n \sim F$$

Null Hypothesis

$$H_0: m = m_0$$

$$W = \sum_{k=1}^{n} \text{signed ranks}$$

Under H_0

CLT (Lyapunov) approximation:

$$Z = \frac{W - 0}{\sqrt{n(n+1)(2n+1)/6}} \sim N(0,1)$$

Note: *W* always differ by 2

(Whole-unit) continuity correction:

$$P(W \geq k) = P(W \geq k-1) = P\left(\underbrace{\frac{W - 0}{\sqrt{n(n+1)(2n+1)/6}}}_{N(0,1)} \geq \frac{k-1}{\sqrt{n(n+1)(2n+1)/6}}\right) = 1 - \Phi\left(\frac{k-1}{\sqrt{n(n+1)(2n+1)/6}}\right)$$

$$P(W \leq k) = P(W \leq k+1) = \Phi\left(\frac{k+1}{\sqrt{n(n+1)(2n+1)/6}}\right)$$

Example

Example 8.4-2

Suppose the lengths of $n = 10$ sunfish are

x_i: 5.0 3.9 5.2 5.5 2.8 6.1 6.4 2.6 1.7 4.3

We shall test H_0: $m = 3.7$ against the alternative hypothesis H_1: $m > 3.7$. Thus, we have

$x_k - m_0$:	1.3,	0.2,	1.5,	1.8,	−0.9,	2.4,	2.7,	−1.1,	−2.0,	0.6
$\lvert x_k - m_0 \rvert$:	1.3,	0.2,	1.5,	1.8,	0.9,	2.4,	2.7,	1.1,	2.0,	0.6
Ranks:	5,	1,	6,	7,	3,	9,	10,	4,	8,	2
Signed Ranks:	5,	1,	6,	7,	−3,	9,	10,	−4,	−8,	2

Therefore, the Wilcoxon statistic is equal to

$$W = 5 + 1 + 6 + 7 - 3 + 9 + 10 - 4 - 8 + 2 = 25.$$

$n = 10$, and so $n(n+1)(2n+1)/6 = 385$. The test is

$$H_0 : m = 3.7, \quad H_1 : m > 3.7.$$

The Wilcoxon signed rank statistic is

$$w = 25.$$

The p-value of the test is

(Whole-unit) continuity correction:

$$P(W \geq 25; H_0) = 1 - P(W \leq 23; H_0) \approx 1 - \Phi\left(\frac{24}{\sqrt{385}}\right) = 0.11,$$

we fail to reject H_0 at 5% significance.

Example

Let m be the median of a symmetric distribution of the continuous type. To test the hypothesis H_0: $m = 160$ against the alternative hypothesis H_1: $m > 160$, we take a random sample of size $n = 16$. For an approximate significance level of $\alpha = 0.05$, H_0 is rejected if the computed $W = w$ is such that

$$z = \frac{w}{\sqrt{16(17)(33)/6}} \geq 1.645,$$

or

$$w \geq 1.645\sqrt{\frac{16(17)(33)}{6}} = 63.626.$$

Say the observed values of a random sample are 176.9, 158.3, 152.1, 158.8, 172.4, 169.8, 159.7, 162.7, 156.6, 174.5, 184.4, 165.2, 147.8, 177.8, 160.1, and 160.5. In Table 8.4-1, the magnitudes of the differences $|x_k - 160|$ have been ordered and ranked. Those differences $x_k - 160$ which were negative have been underlined, and the ranks are under the ordered values. For this set of data,

$$w = 1 - 2 + 3 - 4 - 5 + 6 + \cdots + 16 = 60.$$

fail to reject H_0.

Table 8.4-1 Ordered absolute differences from 160

0.1	0.3	0.5	1.2	1.7	2.7	3.4	5.2
1	2	3	4	5	6	7	8
7.9	9.8	12.2	12.4	14.5	16.9	17.8	24.4
9	10	11	12	13	14	15	16

Example

For the same dataset, say we still want to test

$$H_0: m = 160 \text{ against the alternative hypothesis } H_1: m > 160$$

$$w = 1 - 2 + 3 - 4 - 5 + 6 + \cdots + 16 = 60.$$

What is the p-value of the test?

(Whole-unit) continuity correction:

$$\begin{aligned} p\text{-value} &= P(W \geq 60) = P(W \geq 59) \\ &= P\left(\frac{W - 0}{\sqrt{(16)(17)(33)/6}} \geq \frac{59 - 0}{\sqrt{(16)(17)(33)/6}}\right) \\ &\approx P(Z \geq 1.525) = 0.0636. \end{aligned}$$

$N(0,1)$

- Fail to reject H_0 at the significance level 0.05
- But we will reject H_0 at the significance level 0.1

Extension: data with ties

For continuous distributions, the probability of having two samples X_i and X_j such that $|X_i - m_0| = |X_j - m_0|$ and the probability of having $|X - m_0| = 0$ are both zero. But it could happen in practice.

- If we have $|X_i - m_0| = 0$, simply delete this data and perform the test on remaining samples
- If we have $|X_i - m_0| = |X_j - m_0|$ for some $i \neq j$, then each observation is assigned the average of the corresponding ranks

Example: $H_0: m = 0$ $H_1: m > 0$

1.8	−3.1	0.1	1.1	0.6	−5.1	9.2	0.2	0.4				
0.0	1.9	−0.4	−1.5	1.4	−1.0	2.2	0.8	−0.4				
2.0	−5.8	−3.4	−2.3	3.0	2.7	0.2	3.2					

Raw data

Table 8.4-2 Ordered absolute values, changes in percentage of body fat

0.1	0.2	0.2	0.4	0.4	0.4	0.6	0.8	1.0	1.1	1.4	1.5	1.8
1	2.5	2.5	5	5	5	7	8	9	10	11	12	13
1.9	2.0	2.2	2.3	2.7	3.0	3.1	3.2	3.4	5.1	5.8	9.2	
14	15	16	17	18	19	20	21	22	23	24	25	

Signed ranks: 1, 2.5, 2.5, 5, -5, -5, 7, 8, -9, 10, 11, -12, 13, 14, 15, 16, -17, 18, 19, -20, 21, -22, -23, -24, 25

those that were originally negative underlined

$$w = 1 + 2.5 + 2.5 + 5 - 5 - 5 + \cdots + 25 = 51.$$

Example: data with ties

Since there are $n = 25$ nonzero differences, at $\alpha = 0.05$ significance level, we reject H_0 if

$$z = \frac{w - 0}{\sqrt{(25)(26)(51)/6}} \geq 1.645$$

or, equivalently, if

$$w \geq 1.645\sqrt{\frac{(25)(26)(51)}{6}} = 122.27$$

Since $51 < 122.27$, we fail to reject the null hypothesis. The approximate p-value of this test, using the continuity correction, is

$$p\text{-value} = P(W \geq 51) = P(W \geq 50)$$

(Whole-unit) continuity correction

$$\approx P\left(Z \geq \frac{50 - 0}{\sqrt{(25)(26)(51)/6}}\right) = P(Z \geq 0.673) = 0.2505.$$

Example: data with ties

$$H_0: m = m_0 \qquad H_1: m < m_0$$

Data												
$X_k - m_0$	-2	-2	-1	1	-1	-4	-4	0	2	-5	-3	1
signed rank	-6	-6	-2.5	2.5	-2.5	-9.5	-9.5		6	-11	-8	2.5

delete

- delete the data $x_k - m_0 = 0$, now the sample size is reduced to $n = 11$.
- 4 ties for rank 1,2,3,4: give them average rank $\frac{1+2+3+4}{4} = 2.5$.
- 3 ties for rank 5,6,7: give them average rank $\frac{5+6+7}{3} = 6$.
- 2 ties for rank 9,10: give them average rank $\frac{9+10}{2} = 9.5$.

$$w = -6 - 6 - 2.5 + 2.5 - 2.5 - 9.5 - 9.5 + 6 - 11 - 8 + 2.5 = -44.$$

Not Required

Wilcoxon rank-sum test for two-samples

Reading: Chapter 8.4

Not Required

Wilcoxon rank-sum Test (Mann-Whitney test)

Two samples: Assume F and G have the same shape and spread but possibly different locations, i.e. $G(x) = F(x + c)$

$$X_1, \dots, X_n \sim F$$
$$Y_1, \dots, Y_m \sim G$$

Null Hypothesis

$$H_0: F = G \quad (m_X = m_Y)$$

Alternative Hypothesis

$$H_1: m_X < m_Y$$
$$H_1: m_X > m_Y$$
$$H_1: m_X \neq m_Y$$

Intuition:

1) Rank the two samples together $\{X_1, \dots, X_n, Y_1, \dots, Y_m\}$
2) If $F = G$, the rank of Y are random samples from $\{1,2 \dots, n + m\}$, and each rank is uniform in $\{1,2 \dots, n + m\}$
3) We calculate the sum of ranks for Y, denoted as T_Y, and reject the null hypothesis if the sum is too large or too small.

Let T_Y denote the sum of the ranks of $Y_1, Y_2, \dots, Y_m$.

Not Required

Wilcoxon rank-sum Test (Mann-Whitney test)

Under $H_0: m = m_0$

$$E(T_Y) = \frac{m(m+n+1)}{2}$$

$$\mathrm{Var}(T_Y) = \frac{mn(m+n+1)}{12}$$

When m and n are sufficiently large (>7)

$$\frac{T_Y - E(T_Y)}{\sqrt{Var(T_Y)}} \sim N(0,1)$$

When m and n are small, tables of critical values computed from combinatorial are available.

Not Required

Example

Example 8.4-6 The weights of of the contents of $n_1 = 8$ and $n_2 = 8$ tins of cinnamon packaged by companies A and B, respectively, selected at random, yielded the following observations of X and Y:

x:	117.1	121.3	127.8	121.9	117.4	124.5	119.5	115.1
y:	123.5	125.3	126.5	127.9	122.1	125.6	129.8	117.2

sum of ranks for Y

The critical region for testing $H_0: m_X = m_Y$ against $H_1: m_X < m_Y$ is of the form $w \geq c$. Since $n_1 = n_2 = 8$, at an approximate $\alpha = 0.05$ significance level H_0 is rejected if

$$E(T_Y) = \frac{m(m+n+1)}{2}$$
$$\text{Var}(T_Y) = \frac{mn(m+n+1)}{12}$$

$$z = \frac{w - 8(8+8+1)/2}{\sqrt{[(8)(8)(8+8+1)]/12}} \geq 1.645,$$

or

$$w \geq 1.645\sqrt{\frac{(8)(8)(17)}{12}} + 4(17) = 83.66.$$

Not Required

Example

$$w = 3 + 8 + 9 + 11 + 12 + 13 + 15 + 16 = 87 > 83.66.$$

Reject H_0 at the significance level 0.05

With Half-unit continuity correction

$$\begin{aligned} p\text{-value} &= P(W \geq 87) \\ &= P\left(\frac{W - 68}{\sqrt{90.667}} \geq \frac{86.5 - 68}{\sqrt{90.667}}\right) \\ &\approx P(Z \geq 1.943) = 0.0260. \end{aligned}$$